Nagar YuwakShikshanSanstha's
YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
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Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

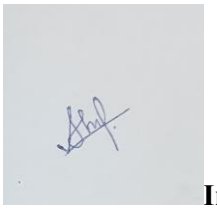
PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.



Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari
04/10/26
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27(ODD)**	**Course Name**	**LAB: Software Engineering**
Semester	**7th**	**Course Code**	**23CT1704**
Roll No	**24**	**Name of Student**	**Srushti Chaudhari**

Practical Number	09
Course Outcome	
Aim	To Perform Manual and Automated testing using CASE tool using Sample GUI
Problem Definition	To Perform Manual and Automated testing using CASE tool using Sample GUI
Theory (100 words)	Manual and automated testing using a CASE (Computer-Aided Software Engineering) tool involves verifying a software application through systematic testing methods. Manual testing is performed by testers who execute test cases on a sample GUI, check inputs and outputs, and identify errors without automation scripts. Automated testing uses testing tools to execute predefined test cases automatically, reducing human effort and increasing accuracy. A sample GUI helps testers verify functions such as buttons, text fields, menus, and navigation. CASE tools support test-case creation, execution, defect tracking, and documentation. These methods improve software quality, reliability, performance, and ensure that the application meets specified requirements.

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Procedure and Execution

(100 Words)

Output:

Code

```
function sum(x, y) { return x+y }
function square(side) { return side * side }
function rectangle(side1, side2) { return side1 * side2; }
function circle(radius) { return Math.PI * radius * radius; }
function right_triangle(base, height) { return 1 / 2 * base * height; }
```

+ Create Test Suite

Title

Summary

+ Add × Cancel

+ Add × Cancel

TS id	Title	Summary	Re
TS0	addition	addition of 2 no.s	
TS1	square	area of sqaure	
TS2	rectangle	side of rectangle	
TS3	circle	circumference of circle	
TS4	triangle	area of traingle	

⊕ Add Test Cases

Test Suite No.	Summary	Script	Expected Output	Actual Output	Manual Testing	
TS0	addition of 2 no.s	sum(12,34)	46	46	Yes	Pass

⊕ Add Test Cases

Test Suite No.	Summary	Script	Expected Output	Actual Output	Manual Testing	
TS0	addition of 2 no.s	sum(12,34)	46	46	Yes	Pass
TS1	area of sqaure	square(5)	25	25	Yes	Pass
TS2	side of rectangle	rectangle(7,4)	28	28	Yes	Pass
TS3	circumference of circle	circle(6)	113	113	Yes	Pass
TS4	area of traingle	triangle(5,9)	22.5	22.5	Yes	Pass

Execute Test Suite

Test Suite: Execution Result

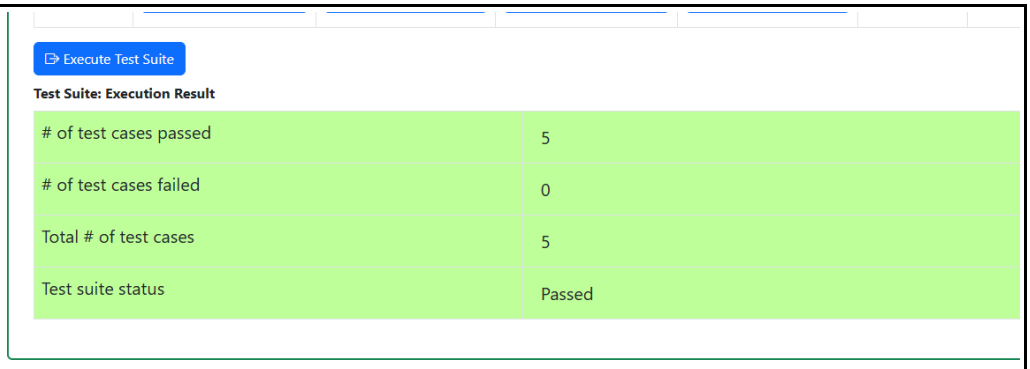
of test cases passed

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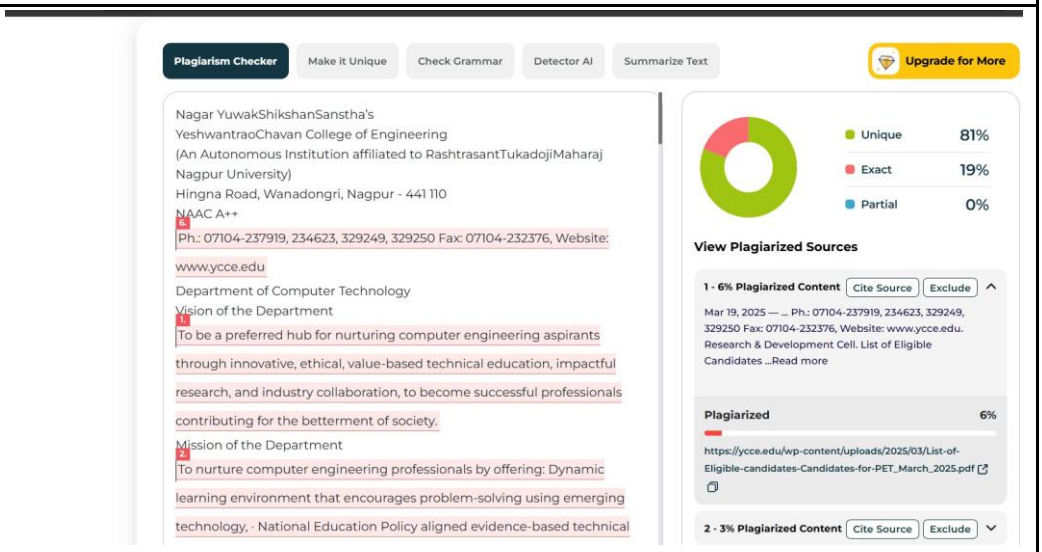
	
Output Analysis	The testing results show that all 5 test cases were executed successfully. The test suite includes addition, square, rectangle, circle, and triangle operations. For each test case, the actual output matches the expected output, so all tests are marked as Pass. 1) TS0: sum(12,34) → Expected: 46, Actual: 46 — Pass 2) TS1: square(5) → Expected: 25, Actual: 25 — Pass 3) TS2: rectangle(7,4) → Expected: 28, Actual: 28 — Pass 4) TS3: circle(6) → Expected: 113, Actual: 113 — Pass 5) TS4: triangle(5,9) → Expected: 22.5, Actual: 22.5 — Pass Final Result: 5 test cases passed, 0 failed, out of 5 total. Therefore, the test suite status is "Passed", indicating that the tested functions are working correctly for the given inputs.
Link of student Github profile where lab assignment has been uploaded	Engineeringhttps://github.com/SRchaudhari15/Software-Engineering-Practicals-24.git
Conclusion	Thus, the practical of **manual and automated testing using a CASE tool** was successfully performed. Five test cases were created and executed for addition, square, rectangle, circle, and triangle operations. The expected outputs matched the actual outputs for all test cases. Hence, **5 test cases passed and 0 failed**, and the overall test suite status was **Passed**. This practical demonstrated how

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	CASE tools can simplify test-case management, execution, and result analysis while improving the reliability and quality of software testing.
Plag Report (Similarity index < 12%)	
Date	04/10/26